

# Dynamics of the Stable Boundary Layer: A Geometric Singular Perturbation Framework for Fold Characterization and Regime Transitions

Part 1: Mathematical Foundations and Observational Resolution

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## Abstract

Observed critical Richardson thresholds vary widely across field campaigns (e.g., CASES-99 and SHEBA), yielding the long-standing Richardson threshold paradox. We show that this variability is not evidence of incompatible physics, but rather the projection of a multidimensional folded critical manifold onto scalar diagnostics. To formalize this mechanism, we construct a multiscale fast-slow model of the stable boundary layer using Geometric Singular Perturbation Theory (GSPT) on a desingularized coordinate chart. The framework yields a sequence of structural results: existence and smoothness of the critical manifold, fold characterization through fast-Jacobian degeneracy, and a constant-rank projection theorem linking invariant state-space geometry to campaign-dependent observational thresholds. Within this geometry, we define an analytical sensible heat-flux turning-capacity limiter,  $H_{\max}$ , and a pathway toward dynamic threshold parameterization. Part 1 establishes the mathematical foundations and observational resolution needed for closure development and data-assimilative validation in Parts 2–3.

# 1 Introduction

The nocturnal Stable Boundary Layer (SBL) remains one of the most persistent challenges in geophysical fluid dynamics due to the highly non-linear, non-equilibrium coupling between turbulent dissipation, surface energy budgets, and mesoscale forcing (Nieuwstadt, 1984; Mahrt, 2014; van de Wiel et al., 2017). A central unresolved problem in boundary-layer meteorology is the so-called “Richardson Number Universality Crisis”: the observation that the critical Bulk Richardson number ( $Ri_c$ ), traditionally expected to be a quasi-universal constant near 0.25, varies wildly across different environmental regimes (Grachev et al., 2013; Monahan et al., 2015). While mid-latitude field campaigns such as CASES-99 report collapse thresholds near the classical limit ( $Ri \approx 0.2$ – $0.4$ ), polar campaigns like SHEBA record active turbulence persisting at stability ratios exceeding unity ( $Ri > 1.2$ ) (Poulos et al., 2002; Grachev et al., 2005).

## 1.1 The Paradox of Stability Thresholds

For decades, the search for a “better” stability function has assumed that the variability in  $Ri_c$  stems from inconsistent physics or measurement noise. However, traditional parameterizations—such as Monin–Obukhov Similarity Theory (MOST) or Mellor–Yamada  $K$ -theory—rely on an instantaneous local equilibrium assumption,  $e = e(S, N^2)$ , that erases the system’s underlying temporal history and structural hysteresis (Mellor and Yamada, 1982). This diagnostic approach forces a smooth, single-valued response that cannot account for the abrupt, “brittle” transitions between weakly stable and very stable regimes.

## 1.2 Thesis: Richardson Thresholds as Projections, Not Invariants

In this paper, we propose that the Richardson threshold paradox is not evidence of incompatible physics, but rather a deterministic consequence of dimensionality reduction. We frame the SBL as a singularly perturbed dynamical system governed by Geometric Singular Perturbation Theory (GSPT) (Fenichel, 1979; Kuehn, 2015). Within this framework, our central thesis is that *Richardson thresholds are low-dimensional coordinate projections of a higher-dimensional folded equilibrium manifold*.

By exploiting the natural separation of timescales—where fast turbulence adjustment ( $\tau_e \sim 10^1$ – $10^2$  s) responds to slow surface thermal and geostrophic evolution ( $\tau_{\text{slow}} \sim 10^3$ – $10^4$  s)—we demonstrate that the “critical” threshold is not a static point, but a state-dependent fold locus ( $\mathcal{C}_{\text{fold}}$ ) that deforms dynamically in response to surface energy balance (van de Wiel et al., 2012).

## 1.3 The “Fold Illusion” and Emergent Coupling

A key conceptual contribution of this work is the distinction between isolated boundary-driven decay and true fold catastrophes. We prove that while an isolated atmospheric subsystem exhibits only a monotonic boundary-crossing at the laminar floor ( $e = 0$ ), a genuine S-shaped fold only emerges through the non-linear coupling of the atmosphere to the Surface Energy Balance (SEB). We term the historical conflation of these two mechanisms the “Fold Illusion,” and show that the geometric resolution of this illusion is necessary to model intermittent bursting and nocturnal relaxation oscillations accurately.

## 1.4 Objectives and Scope of Part 1

As the first in a three-part series, this manuscript establishes the mathematical foundations and observational resolution of the GSPT-SBL framework. We define a hierarchy of models—from 2D analytical baselines to 5D thermodynamic systems—and provide a sequence of structural theorems:

- **Theorem 1 (Existence and Smoothness):** Establishing the critical manifold ( $\mathcal{S}_0$ ) as a smooth embedded submanifold of the state space.
- **Theorem 2 (Fold Characterization):** Defining the geometric boundary where normal hyperbolicity is lost, triggering turbulence collapse.
- **Theorem 3 (Projection Theorem):** Formalizing the mapping from invariant state-space geometry to campaign-specific scalar diagnostics ( $Ri_{\text{fold}}$ ).

Part 1 provides the “geometric skeleton” required for the data-driven discovery and prognostic parameterizations developed in Parts 2 and 3.

- 1.5 The Crisis of Stability Thresholds in Boundary-Layer Meteorology
- 1.6 The Closure Trilemma in Atmospheric Modeling
- 1.7 Objectives and Scope of Part 1

## 2 Multi-Scale Governing Equations and Chart Regularization

This section defines the smooth fast-slow architecture for the stable boundary layer (SBL) model. We introduce the regularized state space, state the standing assumptions used in the geometric analysis, derive the desingularized fast chart, and define the observation map that connects invariant-manifold structure to atmospheric diagnostics.

### 2.1 System Definition and Standing Assumptions

We work with a multiscale autonomous system written in the desingularized time variable  $\tau$  on an open chart domain  $\Omega_0 \subset \mathbb{R}^5$ :

$$\frac{d\mathbf{x}}{d\tau} = \mathbf{F}(\mathbf{x}; \boldsymbol{\mu}, \epsilon_1, \epsilon_2) \equiv \begin{pmatrix} F_{\text{fast}}(\mathbf{x}; \boldsymbol{\mu}) \\ \epsilon_1 F_{\text{slow}}(\mathbf{x}; \boldsymbol{\mu}) \\ \epsilon_1 \epsilon_2 F_{\text{superslow}}(\mathbf{x}; \boldsymbol{\mu}) \end{pmatrix}. \quad (1)$$

The state vector is

$$\mathbf{x} = \begin{pmatrix} \tilde{e} \\ q_\theta \\ S \\ T_s \\ T_g \end{pmatrix} \in \Omega_0 \subset \mathbb{R}_+ \times \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}_+ \times \mathbb{R}_+. \quad (2)$$

Its components are interpreted as follows:  $\tilde{e} \in \mathbb{R}_+$  is the desingularized turbulent kinetic energy coordinate,  $q_\theta \in \mathbb{R}$  is the kinematic turbulent heat flux  $\overline{w'\theta'}$ ,  $S \in \mathbb{R}_+$  is the bulk vertical wind shear  $\|\partial \mathbf{u}/\partial z\|$ ,  $T_s \in \mathbb{R}_+$  is the surface skin temperature, and  $T_g \in \mathbb{R}_+$  is the deep-soil temperature.

**Standing assumptions.** Throughout this section we assume:

1. **(A1) Domain regularity.** The chart domain  $\Omega_0 \subset \mathbb{R}^5$  is open and connected, and the vector field  $\mathbf{F}$  is of class  $C^\infty$  on  $\Omega_0$ .
2. **(A2) Parameter domain.** The parameter block  $\boldsymbol{\mu} \in \mathcal{P}$  belongs to an open bounded set of physically admissible coefficients containing the closure, radiative, and thermodynamic constants appearing below.
3. **(A3) Positive invariance.** The physically admissible set

$$\Omega_{\text{phys}} = \{\mathbf{x} \in \Omega_0 \mid \tilde{e} > 0, S > 0, T_s > 0, T_g > 0\} \quad (3)$$

is positively invariant under the flow.

4. **(A4) Local well-posedness.** For each initial state  $\mathbf{x}(0) \in \Omega_0$ , there exists a unique local trajectory  $\mathbf{x}(\tau)$ .
5. **(A5) Timescale separation.** The perturbation parameters satisfy

$$0 < \epsilon_1 \ll 1, \quad 0 < \epsilon_2 \ll 1, \quad (4)$$

with fast dynamics on  $O(1)$  time, slow dynamics on  $O(\epsilon_1^{-1})$  time, and super-slow dynamics on  $O((\epsilon_1 \epsilon_2)^{-1})$  time.

Unless otherwise stated, we use  $\Omega_0$  to denote the full mathematical chart domain and  $\Omega_{\text{phys}}$  to denote the positively invariant physical region used for geometric analysis. We further assume that  $\Omega_0$  extends across  $\partial\Omega_{\text{phys}}$  to include a neighborhood of  $\tilde{e} = 0$ , so that  $\mathcal{S}_0$  and  $\mathcal{C}_{\text{fold}}$  can be analyzed as smooth embedded submanifolds near the laminar boundary.

## 2.2 Chart Regularization and Desingularization

In the original turbulent kinetic energy variable  $e$ , the fast equations contain non-smooth fractional powers through the shear-production term  $c_m \ell e^{1/2} S^2$  and the Kolmogorov dissipation term  $e^{3/2}/\ell$ . As  $e \rightarrow 0^+$ , differentiability is lost at the laminar boundary.

To restore smoothness, introduce the regularized chart map

$$\phi_\delta : e \mapsto \tilde{e} = \sqrt{e + \delta}, \quad \delta > 0. \quad (5)$$

We formulate the geometric analysis on the regularized chart and use the limit  $\delta \rightarrow 0^+$  as the asymptotic interpretation of the physical system. To desingularize the fast dynamics, use the positive time reparameterization

$$d\tau = \frac{dt}{\epsilon_1 \tilde{e}} \quad \Longleftrightarrow \quad \frac{dt}{d\tau} = \epsilon_1 \tilde{e}. \quad (6)$$

**Proposition 1** (Regularization and diffeomorphic equivalence). *On the physical domain  $e > 0$ , the map  $\phi_\delta$  is a  $C^\infty$  diffeomorphism onto its image. Under the positive time change*

$$d\tau = \frac{dt}{\epsilon_1 \tilde{e}}, \quad (7)$$

*the fast subsystem extends to a smooth vector field in the chart variable  $\tilde{e}$ , while preserving phase trajectories, equilibrium sets, and the signs of eigenvalues transverse to normally hyperbolic branches of the critical manifold.*

*Proof.* For  $\delta > 0$  and  $e > 0$ ,

$$\phi_\delta(e) = \sqrt{e + \delta} \quad (8)$$

is smooth with derivative

$$\phi'_\delta(e) = \frac{1}{2\sqrt{e + \delta}} > 0, \quad (9)$$

so  $\phi_\delta$  is a smooth diffeomorphism onto its image. In the physical variable  $e$ , the fast TKE equation is

$$\epsilon_1 \frac{de}{dt} = c_m \ell e^{1/2} S^2 - \frac{g}{\theta_0} q_\theta - \frac{1}{\ell} e^{3/2}. \quad (10)$$

Since  $\tilde{e}^2 = e + \delta$ , differentiation gives

$$2\tilde{e} \frac{d\tilde{e}}{dt} = \frac{de}{dt}, \quad (11)$$

and hence

$$\frac{d\tilde{e}}{d\tau} = \frac{dt}{d\tau} \frac{d\tilde{e}}{dt} = \epsilon_1 \tilde{e} \cdot \frac{1}{2\tilde{e}} \frac{1}{\epsilon_1} \frac{de}{dt} = \frac{1}{2} \frac{de}{dt}. \quad (12)$$

Substituting the physical equation and using  $e = \tilde{e}^2 - \delta$  yields

$$\frac{d\tilde{e}}{d\tau} = \frac{1}{2} \left( c_m \ell \sqrt{\tilde{e}^2 - \delta} S^2 - \frac{g}{\theta_0} q_\theta - \frac{1}{\ell} (\tilde{e}^2 - \delta)^{3/2} \right). \quad (13)$$

In the limiting chart  $\delta \rightarrow 0^+$  this becomes

$$\frac{d\tilde{e}}{d\tau} = \frac{1}{2}c_m\ell S^2\tilde{e} - \frac{g}{2\theta_0}q_\theta - \frac{1}{2\ell}\tilde{e}^3, \quad (14)$$

which is polynomial in  $\tilde{e}$ . Because  $d\tau/dt = \tilde{e}/\epsilon_1 > 0$  on  $\Omega_{\text{phys}}$ , the reparameterization is orientation preserving. Standard results for positive time changes then imply preservation of orbits, equilibria, and the signs of eigenvalues transverse to normally hyperbolic branches.  $\square$

### 2.3 Subsystem Vector Field Components

Under Assumption (A1) and Proposition 2.1, the desingularized vector field components are given explicitly as follows.

**Fast subsystem.** The fast subsystem  $F_{\text{fast}}(\mathbf{x}) = (\tilde{F}(\mathbf{x}), \tilde{H}(\mathbf{x}))^T$  governs rapid turbulent adjustment:

$$F_{\text{fast}}(\mathbf{x}) = \begin{pmatrix} \frac{1}{2}c_m\ell S^2\tilde{e} - \frac{g}{2\theta_0}q_\theta - \frac{1}{2\ell}\tilde{e}^3 \\ -c_w\theta_z(T_s)\tilde{e}^3 - \frac{g}{\theta_0}c_\theta\ell q_\theta^2 - \frac{C_\theta}{\ell}\tilde{e}^2q_\theta \end{pmatrix}. \quad (15)$$

Here the local thermal stratification is represented by the smooth diagnostic map

$$\theta_z(T_s) = \frac{\theta_{\text{top}} - \Pi_s T_s}{\Delta z}. \quad (16)$$

**Slow subsystem.** The slow subsystem  $F_{\text{slow}}(\mathbf{x}) = (F_S(\mathbf{x}), F_{T_s}(\mathbf{x}))^T$  governs wind shear and surface thermal evolution:

$$F_{\text{slow}}(\mathbf{x}) = \begin{pmatrix} \tilde{e} \left[ \mathcal{F}_{\text{ls}} - \frac{\partial}{\partial z}(c_m\ell\tilde{e}S) \right] \\ \frac{\tilde{e}}{C_s} \left[ R_{\text{net}}(T_s) + \rho c_p q_\theta + \frac{k_g}{d_g}(T_g - T_s) \right] \end{pmatrix}, \quad (17)$$

where the factors  $\tilde{e}$  arise from the positive time reparameterization, so the slow tendencies freeze in fast time as  $\tilde{e} \rightarrow 0^+$ . The corresponding net radiation closure is

$$R_{\text{net}}(T_s) = R_{\text{sw}}^\downarrow + \epsilon_a \sigma T_a^4 - \epsilon_s \sigma T_s^4. \quad (18)$$

**Super-slow subsystem.** The super-slow subsystem governs deep-soil heat conduction:

$$F_{\text{superslow}}(\mathbf{x}) = \tilde{e} \frac{k_g}{d_g^2}(T_s - T_g). \quad (19)$$

As in the slow subsystem, the prefactor  $\tilde{e}$  is induced by the positive time change; all geometric arguments in subsequent sections are restricted to  $\Omega_{\text{phys}}$ .

### 2.4 Observation Mapping and Diagnostic Functionals

To connect the internal state  $\mathbf{x} \in \Omega_0$  to observable diagnostics, define the smooth observation map

$$\mathbf{\Pi}_{\text{obs}} : \Omega_0 \rightarrow \mathcal{O} \subset \mathbb{R}^3, \quad (20)$$

where  $\mathcal{O}$  denotes the space of smooth diagnostic quantities accessible from field observations or numerical model output. Its components are

$$\mathbf{\Pi}_{\text{obs}}(\mathbf{x}) = \begin{pmatrix} \pi_{Ri}(\mathbf{x}) \\ \pi_H(\mathbf{x}) \\ \pi_S(\mathbf{x}) \end{pmatrix} = \begin{pmatrix} \frac{g}{\theta_0} \frac{\theta_z(T_s)}{S^2} \\ \rho c_p q_\theta \\ S \end{pmatrix}. \quad (21)$$

These functionals represent the gradient Richardson diagnostic, the sensible heat flux, and the wind-shear coordinate, respectively.

**Differential structure of  $\pi_{Ri}$ .** The differential covector of the Richardson functional is

$$D\pi_{Ri}(\mathbf{x}) = \begin{pmatrix} 0 & 0 & -\frac{2g\theta_z(T_s)}{\theta_0 S^3} & \frac{g}{\theta_0 S^2} \frac{\partial \theta_z}{\partial T_s} & 0 \end{pmatrix}. \quad (22)$$

Hence  $D\pi_{Ri}$  lies entirely in the slow cotangent subspace  $\text{span}\{dS, dT_s\}$ , so the Richardson diagnostic is insensitive, at first order, to perturbations in the fast turbulent directions  $(\tilde{e}, q_\theta)$ .

**Proposition 2** (Observation smoothness). *Under Assumptions (A1) and (A3), the map  $\mathbf{\Pi}_{\text{obs}}$  and its component functionals  $(\pi_{Ri}, \pi_H, \pi_S)$  are  $C^\infty$  on  $\Omega_0$ .*

*Proof.* Each component of  $\mathbf{\Pi}_{\text{obs}}$  is a composition of smooth algebraic operations on  $\Omega_0$ . Because  $S > 0$  on  $\Omega_{\text{phys}}$ , the denominator in

$$\pi_{Ri}(\mathbf{x}) = \frac{g}{\theta_0} \frac{\theta_z(T_s)}{S^2} \quad (23)$$

never vanishes. The remaining components are linear in  $q_\theta$  and  $S$ , and  $\theta_z(T_s)$  is smooth by construction. Therefore  $\mathbf{\Pi}_{\text{obs}} \in C^\infty(\Omega_0, \mathcal{O})$ .  $\square$

**Remark 2.2 (Geometry versus observation).** The dynamical system evolves on the five-dimensional state manifold  $\Omega_0$ , whereas atmospheric diagnostics are represented in observation space through the smooth map

$$\mathbf{\Pi}_{\text{obs}} : \Omega_0 \rightarrow \mathcal{O}. \quad (24)$$

Consequently, invariant geometric objects in state space need not appear as invariant objects in observation space.

Section 2 now provides a smooth desingularized dynamical system, an explicit regularized chart, and a diagnostic observation framework suitable for the invariant-manifold results developed in the subsequent sections.



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